EDINBURGH NAPIER UNIVERSITY

School of Computing, Engineering and the Built Environment

PhD PROGRESS REPORT - RD6

April 2026 - October 2026

Thesis Title:

Development of Novel Artificial Intelligence Models for Early Detection and Prediction of Infectious Diseases with a Focus on Sepsis

Field	Detail
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Reporting period	April 2026 - October 2026
Report type	RD6 - Six-Monthly Progress Review
Status of results in this report	Preliminary, development-split results on 10,000-stay samples; the full-cohort results are being prepared for journal submission; the sealed holdout has not been opened.

1. Introduction

Sepsis, a dysregulated host response to infection that leads to organ dysfunction, affects about 49 million people a year and causes about 11 million deaths. Each hour of delay in antibiotic treatment raises mortality by roughly 4 to 8 per cent, yet the clinical definition (Sepsis-3: suspected infection plus a rise of at least two points on the SOFA organ-failure score) is met only once organ damage has begun. An early warning that arrives hours before the definition is met, and that clinicians trust enough to act on, remains the unmet need this PhD addresses.

Published sepsis prediction models report excellent discrimination on their own data and fail where it matters. The winning entry of the PhysioNet 2019 Challenge scored a normalised utility of 0.36 on the two hospitals it trained on and -0.12 on a hidden third hospital, worse than never alarming (Reyna et al., 2020). The Epic Sepsis Model, deployed in hundreds of US hospitals, achieved an AUROC of 0.63 in independent validation against a vendor claim of 0.76 to 0.83, missing 67 per cent of sepsis patients while alerting on 18 per cent of all admissions (Wong et al., 2021). Three causes recur: information from the future leaks into training; models are scored by ranking rather than by whether an alarm would help; and every study defines sepsis its own way.

This reporting period therefore re-oriented the programme. An audit of the candidate's own earlier pipeline found look-ahead leakage; those models were retired and their results are no longer reported. The work moved from the challenge's prepared tables to two raw ICU databases, MIMIC-IV 3.1 and eICU-CRD 2.0, with one shared Sepsis-3 definition, strict hour-by-hour causality, a clinician-defined predictor set, and evaluation by clinical utility with a sealed holdout. The research design became mixed-methods: the quantitative modelling now under way, a qualitative study with the ICU

clinicians who would receive the alarm, and, as the future direction, a prospective test on hospital patients.

2. Research Aim and Objectives

The overarching aim is unchanged: to develop and evaluate novel AI models that enable early, accurate and interpretable detection and prediction of infectious diseases, with an emphasis on sepsis in the ICU. The central research question, refined this period, is:

Using only the information a clinician could have seen at each hour, can a model warn of sepsis six hours ahead with positive clinical utility, and does that utility survive transfer to a hospital system it has never seen?

2.1 Objectives

1. Build a causal, cross-database pipeline: one Sepsis-3 definition applied identically to MIMIC-IV and eICU, every value used only from the hour it was recorded, hours after onset excluded, and a 20 per cent patient holdout sealed until the protocol is frozen.
2. Define the predictor set from the bedside: the full clinical picture (655 predictors), with antibiotics and cultures that define the label excluded, prophylactic antibiotics separated by a guideline rule, and clinician-suspicion proxies declared and tested in sensitivity analyses.
3. Evaluate architectures as hypotheses, including the candidate's OSTA-Transformer, against the same model without organ structure, simpler sequence models, the PhysioNet winner's method and self-supervised pretraining, with decision rules fixed before results exist.
4. Judge models by what would help patients: normalised utility, sensitivity at a fixed false-alarm burden, warning time, calibration and transfer between databases, with patient-level confidence intervals.
5. Conduct a qualitative study with ICU doctors, nurses and pharmacists on whether they would act on the alarm, what explanation they need, and what false-alarm rate they would accept.
6. Prepare a prospective, initially silent, test of the model on hospital patients, and disseminate the work through peer-reviewed publication.

2.2 Research questions

- **RQ1 Predictors.** Which information known at hour t carries the signal: vital signs and laboratory values alone, or also treatments, devices, admission context and comorbidity that no public model uses?
- **RQ2 Architecture.** Once causality is enforced, does structuring the model around organ systems add anything over simpler sequence models?
- **RQ3 Transfer and trust.** Does positive utility in one ICU database survive in another, and what would make the clinicians who receive the alarm act on it?

3. Research Methodology

3.1 Mixed-methods design

Phase	Method	Output
Phase 1 - Quantitative (in progress)	Retrospective modelling on MIMIC-IV and eICU: one label definition, causal features, prespecified architecture and loss comparison, utility-based evaluation, transfer between databases, sealed holdout opened once.	A warning model with measured utility and an alarm policy.
Phase 2 - Qualitative (planned, 2027)	Semi-structured interviews with patient vignettes and a survey built on the model's own alarms, with ICU doctors, nurses and pharmacists; thematic analysis; ethics approval to be sought.	The false-alarm budget, the explanation (overall risk or risk per organ) and the trust conditions the model must meet.
Phase 3 - Prospective (future direction)	Silent-mode run on live ICU data with alarms logged but not shown, compared with outcomes; then a clinician-facing pilot. Requires ethics approval, a data-governance agreement and a clinical partner.	Evidence from real patients, in real time, in a hospital the model never trained on.

The design is mixed because an alarm that clinicians do not trust saves nobody: the qualitative phase sets the operating constraints that the quantitative model must satisfy before any prospective use.

3.2 Data sources

Database	Setting	Candidate ICU stays	Used this period
MIMIC-IV 3.1	Beth Israel Deaconess Medical Center, Boston (one academic centre)	about 94,000	fixed random sample of 10,000 stays; 8,795 after benchmark exclusions
eICU-CRD 2.0	about 200 US hospitals (Philips eICU programme)	about 201,000	fixed random sample of 10,000 stays; 5,710 after benchmark exclusions
PhysioNet 2019	two US hospitals, challenge-prepared hourly tables	40,336 patients	causal re-evaluation of the earlier architectures only

Both raw databases are accessed under PhysioNet credentialing. Patient-level data never leave the analysis environment; only aggregate results are reported.

3.3 Labels, cohorts and splits

- Sepsis-3 onset is derived with the published Yet Another ICU Benchmark (YAIB) pipeline on the ricu concept library, identically for both databases: suspected infection (an antibiotic order and a culture within the Seymour window in MIMIC-IV; antibiotics only in eICU, where cultures are not reliably recorded) plus a SOFA rise of at least two points.

- The prediction task is hourly: does onset fall within the next six hours? Hours after onset are excluded from training and scoring; stays are followed for up to 120 hours.
- 20 per cent of patients in each database are sealed as the final holdout by a deterministic patient hash. The remaining 80 per cent form the development set, split by patient into training, validation, calibration and test. No threshold, calibration or model choice ever touches the holdout.
- The 10,000-stay samples are fixed random draws, identical across every run this period, so differences between runs are due to the methods.

3.4 Causality and predictors

Every value is used only from the hour it was recorded in the system (store time where available), expires after 24 hours (no expiry for weight and Glasgow Coma Scale components), and is accompanied by a measured-or-not flag and an age channel. The 655 predictors were defined by the candidate from clinical practice and fall into the following groups:

Group	Content
Vital signs and monitoring	heart rate, cuff and arterial blood pressures, respiratory rate, oxygen saturation, temperature, end-tidal CO2, central venous pressure, cardiac output, pain and sedation scores
Laboratory and blood gas	chemistry, haematology with differential, coagulation, lactate, procalcitonin, troponin, BNP, ferritin, D-dimer, amylase, lipase, ammonia, urinalysis, blood gases
Scores and derived	SOFA (total and per organ), qSOFA, NEWS2, shock index, P/F and S/F ratios, neutrophil-lymphocyte ratio, BMI, changes against the previous value and the baseline, 24-hour extremes and trends
Organ support	vasopressor dose in noradrenaline equivalents, ventilation mode and settings, oxygen flow, airway status, renal replacement therapy, urine output rate and KDIGO AKI stage
Fluids, drugs and devices	crystalloids and transfusions, nutrition route, central and arterial lines, urinary catheters, drains; corticosteroids, immunosuppressants, chemotherapy, sedatives, opioids, diuretics, antipyretics
Context and history	admission source and ICU type, hours in hospital before ICU, prior admissions, prior positive cultures and antibiotics in earlier admissions, 17 comorbidity flags, infection-related phenotypes, imaging orders
Prophylactic antibiotics (MIMIC-IV)	surgical prophylaxis identified by agent, perioperative timing, duration of 24 hours or less (48 for cardiac surgery), no culture in the suspicion window and no escalation within 72 hours (ASHP/IDSA/SIS/SHEA 2013; Surviving Sepsis Campaign 2021); medical prophylaxis by agent and dose pattern
Never used as predictors	antibiotics and cultures that define suspected infection; race, ethnicity and insurance (fairness analysis only); anything recorded after the prediction hour

Two inputs are flagged as clinician-suspicion proxies, stress-dose hydrocortisone and imaging orders, because they often encode that the team already suspects sepsis. They are kept, declared, and every result is reported with and without them.

3.5 Models and the prespecified comparison

- Kept from earlier runs: OSTA-Transformer (organ-system-aware temporal attention with cross-organ attention and per-organ risk), output fusion, CNN-Transformer, logistic regression and boosted trees.
- Literature benchmark: path signatures with boosted trees, the method of the PhysioNet 2019 winner, replicated on the same hours.
- New supervised models: OSTA without organ grouping (the ablation that decides whether organ structure is kept), GRU-D, a causal temporal CNN, a CNN encoder with an OSTA head, and event attention with an OSTA head.
- Self-supervised: a per-channel causal CNN encoder pretrained on up to 75,000 unlabelled stays per database (next-value forecasting, measured-within-six-hours, SOFA organ components and organ worsening at 6, 12 and 24 hours), fine-tuned with OSTA, flat and discrete-time survival heads; site-specific and pooled encoders.
- Losses: utility-weighted cross-entropy (primary), plain cross-entropy and temporal label smoothing, chosen by the rank-average of validation utility and AUPRC.
- Decision rules written before any result: organ structure is kept only if OSTA beats its flat counterpart on validation; the shortlist for the full cohort is the top two or three new models plus the baselines; the full cohort uses three seeds.

3.6 Evaluation

The principal panel is the PhysioNet normalised utility (official scorer, unchanged), hourly AUROC and AUPRC, sensitivity at a budget of 0.5 false alarms per patient-day, median warning time before onset, false alarms per patient-day, and calibration (Brier score, calibration slope). Additions this period: warning time relative to the first antibiotic order, transfer rows (each database's model scored on the other's development patients), a conformal alarm policy with abstention on out-of-distribution hours, a trajectory-rich subset, and the no-proxy sensitivity analysis. All intervals are patient-cluster bootstraps; paired comparisons against OSTA use the same bootstrap.

3.7 Engineering and reproducibility

The pipeline is a set of self-contained Colab notebooks with all analysis code embedded and verified by SHA-256 checksums, 212 automated engineering checks (including future-perturbation invariance tests for every model), resumable stages, and a synthetic-data fixture that runs the whole protocol end to end. Nothing calls an external AI service and patient data never leave the environment. Compute this period was Google Colab with an A100 GPU and high-memory runtimes.

4. Research Progress (April - October 2026)

4.1 Leakage audit and retirement of the earlier models

A line-by-line audit of the PhysioNet pipeline used for the models described in the March 2026 report found three kinds of look-ahead leakage: hours after onset were labelled positive, measurements were carried forward from the time they were taken rather than the time they became available, and treatments that define the label were among the inputs. Those models and their reported metrics have been withdrawn and are not cited in this report; the two manuscripts based on them were withheld from submission.

Rebuilt causally, the OSTA-Transformer on the PhysioNet 2019 test set (three-seed ensemble) achieves a patient-level AUROC of 0.81, an hourly AUROC of 0.80 and a normalised utility of 0.22

(0.30 at the conformal operating point, sensitivity 0.70, specificity 0.72). With 1.6 per cent of hours positive, these are the honest baseline figures from which all later work is measured. This finding is itself a contribution: it demonstrates how far leakage inflates published scores and defines the audit that the new pipeline passes by construction.

4.2 The causal cross-database pipeline

Sepsis-3 labels and cohorts were extracted for both raw databases with the benchmark pipeline, including a repair of a defect in the pinned benchmark code that halted eICU extraction. The feature pipeline passed through five versions during the period, each adding predictor classes identified from clinical practice (comorbidity and admission context, devices and drug exposures, vasopressor dose, full SOFA and lab deltas, prophylactic antibiotics), to the current 655 predictors with masks and ages. All stages are keyed and resumable, and the raw-table scans are shared between runs.

4.3 Extraction audit

A coverage audit against the raw tables showed that much of the apparent transfer problem lies in how data are charted. Values that most published pipelines read from the periodic monitoring tables were, in eICU, largely in the nurses' charting table; urine output in MIMIC-IV became a rate per kilogram per hour only once each record was tied to the interval it covered. Coverage after the audit (share of patient-hours with a usable value):

Channel	Before audit	After audit
eICU temperature	5 %	94 %
eICU blood pressure (cuff and arterial)	21 %	99.5 %
eICU weight	6 %	96 %
eICU Glasgow Coma Scale	62 %	83 %
MIMIC-IV urine output rate	9 %	44 %
MIMIC-IV Glasgow Coma Scale	66 %	78 %

4.4 Preliminary results on the development split

The following are preliminary results on the development test split of the 10,000-stay samples (one to three seeds). The sealed holdout has not been opened and nothing has been tuned on test data. Full results will appear in the manuscript in preparation.

Measure	MIMIC-IV development test split
Test split	1,422 stays, 63,488 hours, 71 sepsis events; 0.65 % of hours positive
Hourly AUROC	0.81 (three seeds, SD 0.012) for OSTA and the CNN-Transformer; 0.80 output fusion; 0.78 logistic regression; 0.76 boosted trees
Hourly AUPRC	0.03 to 0.04 at 0.65 % prevalence (chance 0.0065)
Sensitivity at 0.5 false alarms per patient-day	about one sepsis event in two
Median warning before onset	5 hours

Measure	MIMIC-IV development test split
Normalised utility	best model 0.18 (95 % CI 0.08 to 0.27); OSTA 0.12
Reference points	PhysioNet 2019 winner: 0.36 on its training hospitals, -0.12 on the hidden hospital; our causal OSTA on PhysioNet: 0.22

The utility on MIMIC-IV is modest but clearly positive on a database built from raw tables with every hour scored causally, whereas the best competition entry fell below zero as soon as it left its training hospitals. On eICU the development-split AUROC is 0.67 to 0.69 and the utility is not yet distinguishable from zero; eICU has the weaker antibiotic-only label and, before the audit, the poorest coverage, and it is the focus of the full-cohort, leave-hospital-out and label-harmonisation work planned below. The candidate's replication of the winner's own method on both databases is in progress.

4.5 Overfitting and the architecture comparison

Training records show that every rich-feature neural model reaches its best validation score within one to three epochs while the training loss keeps falling: a training split contains about 175 septic stays, too few for 300 to 650 predictors. The AUROCs of OSTA, the CNN-Transformer, output fusion and the gated variants differ by less than the seed-to-seed noise, and OSTA with and without attention gates gives identical results because the chosen checkpoint is the first epoch either way. The conclusion is that on this sample size nothing separates the architectures, and that the next lever is data: the full cohort (about nine times more sepsis cases), self-supervised pretraining on unlabelled stays, and a training loss chosen by utility.

The prespecified comparison of sixteen models, three losses and the organ-structure ablation, with self-supervised pretraining and the winner's method, is running at the time of writing; its shortlist, chosen by the validation rule, goes to the full-cohort run.

4.6 Publications and dissemination

Three manuscripts were submitted during the period (Section 5). The main study on MIMIC-IV and eICU is being written up for submission in 2027 once the full-cohort and sealed-holdout results are available. The candidate presents the work at the SCEBE Research Degree Conference on 20 October 2026 (15-minute talk and questions).

4.7 Deviations from the March 2026 plan

Planned in March 2026	What happened	Reason
Submit the CNN-Transformer paper (Apr-May) and the OSTA paper (May-Jun)	Withheld	leakage audit; both architectures are being re-evaluated causally in the current comparison
Respond to PLOS ONE reviewer feedback on the AMR review	Revised and submitted to the International Journal of Antimicrobial Agents (9 Oct 2026)	change of venue after revision
External validation of OSTA on MIMIC-IV (Jun-Aug)	Became the core programme: MIMIC-IV and eICU, one label, causal pipeline, utility evaluation	validation on prepared data would have inherited the leakage

Planned in March 2026	What happened	Reason
Real-time deployment feasibility	Deferred to the prospective phase	depends on the full-cohort model and the clinician study
Clinical stakeholder study with ICU clinicians	Now the qualitative phase of a mixed-methods design (Feb-Apr 2027)	sequenced after the full-cohort results so that the alarms shown to clinicians are real
Conference presentation	SCEBE Research Degree Conference, 20 Oct 2026	as planned

5. Publications Summary

#	Title	Journal	Submitted	Status
1	Artificial Intelligence in the Diagnosis of Antimicrobial Resistance: Rapid Diagnostics and Imaging Innovations (narrative review)	International Journal of Antimicrobial Agents	9 Oct 2026	Technical check
2	Temporal Patterns and Predictors of Sepsis Onset in ICU Patients: A Retrospective Cohort Study Using PhysioNet Data	Frontiers in Medicine	23 Sep 2026	Editorial assignment
3	Diagnostic Applications of Infrared Thermography Across Infectious Diseases: A Systematic Review with a Subgroup Meta-Analysis	JMIR	18 Jul 2026	In review
4	Early sepsis warning on MIMIC-IV and eICU: causal predictors, clinical utility and transfer (main study)	target: critical care informatics journal	planned 2027	In preparation

The cohort study (manuscript 2; 27,101 ICU admissions, 8 per cent sepsis, median onset 100 hours) is the statistical companion to the modelling work: Cox models with a six-hour backshifted label identify early predictors that the deep models then use.

6. Professional Development

Activity	Status	Relevance to the PhD
Harvard PPCR: Clinical Trials Design, Implementation and Analysis (nine-month programme)	Completed, February 2026	informs the alarm-policy study, the qualitative phase and the design of the prospective test
Iowa State University: Machine Learning Programme	Completed, December 2025	theoretical grounding for the architecture comparison and model selection
DataCamp Python Data Science Career Track	In progress since September 2025	data engineering for the raw-database pipeline

Activity	Status	Relevance to the PhD
PhysioNet credentialed access to MIMIC-IV 3.1 and eICU-CRD 2.0	Held	all empirical work this period
SCEBE Research Degree Conference, 15-minute presentation	20 October 2026	presentation skills; feedback on model selection and the clinician study design

Skills developed this period: large-scale extraction from raw ICU databases (DuckDB over multi-gigabyte tables; item dictionaries and unit audits), the ricu and YAIB benchmark tooling, causal and availability-aware feature design, utility-based evaluation and the official PhysioNet scorer, prespecified analysis protocols with sealed holdouts, reproducible notebook engineering with automated checks, and GPU training on cloud runtimes. The candidate also led the clinical specification of the predictor set, the antibiotic classification rule and the proxy analysis, which is where a pharmacist's training directly shaped the method.

7. Plan for the Next Six Months (November 2026 - April 2027)

When	Objective	Deliverable
Oct - Nov 2026	Complete the sixteen-model comparison; apply the prespecified validation rules (loss, organ structure, shortlist)	shortlist and rules recorded; comparison section of the manuscript
Nov - Dec 2026	Full cohort (about 94,000 MIMIC-IV and 201,000 eICU stays), three seeds; leave-hospital-out evaluation across eICU hospitals; pooled multi-site training; label-harmonisation sensitivity (antibiotic-only suspicion on MIMIC-IV)	full-cohort development results; compute plan (Colab or a dedicated cloud machine)
Jan 2027	Freeze the protocol; open the sealed holdout once; fix the alarm policy (sensitivity against false alarms per patient-day); subgroup fairness analysis	holdout results; alarm policy; manuscript draft
Feb - Apr 2027	Qualitative study: ethics application, interviews with vignettes and a survey using the model's own alarms, with ICU doctors, nurses and pharmacists; thematic analysis	ethics approval; interview guide and survey; findings on trust, explanation and alarm burden
Feb - Apr 2027	Radiology reports as predictors (MIMIC-IV notes, rule-based findings with local text models; no external services)	ablation with and without notes
From mid-2027	Future direction: prepare the silent prospective test on hospital patients (clinical partner, data-governance agreement, ethics)	protocol for the prospective phase; submission of the main study

7.1 Risks and mitigations

- **Compute time.** The full cohort is about nine times the current sample; Colab runs of 12 to 16 hours per experiment are at the limit. Mitigation: a dedicated cloud GPU machine for the full-cohort runs, with the same resumable pipeline.

- **Label heterogeneity.** eICU's antibiotic-only suspicion is weaker than MIMIC-IV's. Mitigation: report transfer under both definitions and a harmonised antibiotic-only sensitivity analysis.
- **Small-sample overfitting.** Mitigation: scale, self-supervised pretraining, three seeds, and a shortlist chosen on validation rather than test.
- **Ethics lead time for the qualitative study and the prospective phase.** Mitigation: start the ethics application in January 2027 and secure a clinical partner during the qualitative phase.

8. Conclusion

The period turned a set of promising but leaky results into an honest, transferable research programme. The candidate audited and retired the earlier models, built a causal pipeline across two raw ICU databases with one shared definition of sepsis and a clinician-defined predictor set, moved evaluation to clinical utility with a sealed holdout, and produced preliminary evidence that a causally scored warning keeps positive utility on MIMIC-IV where the best published competition entry collapsed on an unseen hospital. Three manuscripts were submitted. The next period scales the evaluation to the full cohort, opens the holdout once, brings in the clinicians who would receive the alarm, and prepares the prospective test on hospital patients that the whole programme points towards.